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ZoomEarth: Active Perception for Ultra-High-Resolution Geospatial Vision-Language Tasks

Ruixun Liu[1*], Bowen Fu[1*], Jiayi Song[1], Kaiyu Li[1], Wanchen Li[1], Lanxuan Xue[1],
Hui Qiao[3], Weizhan Zhang[1], Deyu Meng[2,4,5], Xiangyong Cao[1,2†]
[1]Faculty of Electronic and Information Engineering, Xi'an Jiaotong University
[2]Ministry of Education Key Laboratory of Intelligent Networks and Network Security
[3]China Telecom Shaanxi Branch [4]School of Mathematics and Statistics, Xi'an Jiaotong University
[5]Macao Institute of Systems Engineering, Macau University of Science and Technology

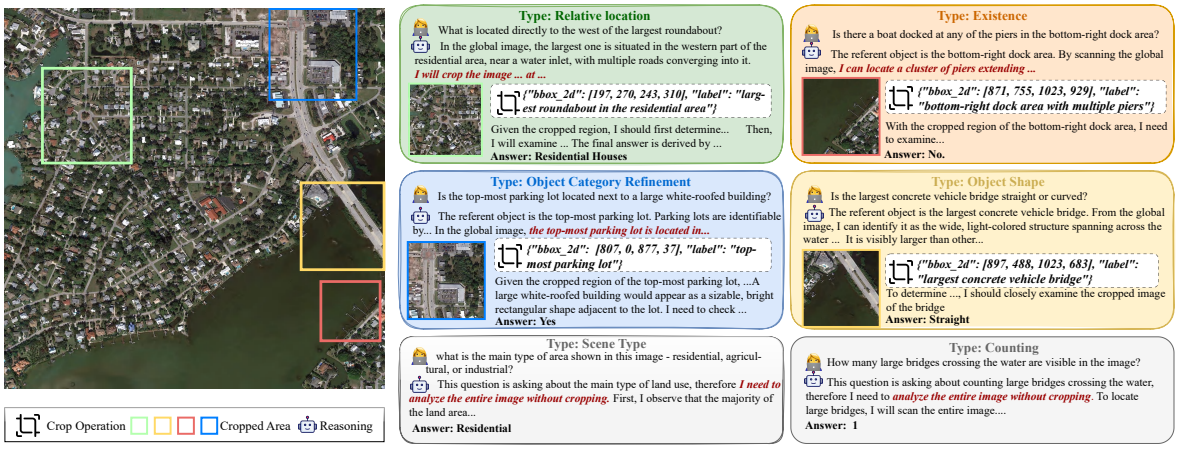


Figure 1. Typical examples from our proposed benchmark LRS-GRO and results obtained by our ZoomEarth framework. LRS-GRO focuses on UHR RS imagery, including 17 multimodal vision-language understanding categories and emphasizing active perception process.

Abstract

Ultra-high-resolution (UHR) remote sensing (RS) images offer rich fine-grained information but also present challenges in effective processing. Existing dynamic resolution and token pruning methods are constrained by a passive perception paradigm, suffering from increased redundancy when obtaining finer visual inputs. In this work, we explore a new active perception paradigm that enables models to revisit information-rich regions. First, we present LRS-GRO, a large-scale benchmark dataset tailored for active perception in UHR RS processing, encompassing 17 question types across global, region, and object levels, annotated via a semi-automatic pipeline. Building on LRS-GRO, we propose ZoomEarth, an adaptive cropping–zooming framework with a novel Region-Guided reward that provides fine-grained guidance. Trained via supervised fine-tuning (SFT) and Group Relative Policy Optimization (GRPO), ZoomEarth achieves state-of-the-art performance on LRS-GRO and, in the zero-shot setting, on three public UHR remote sensing benchmarks. Furthermore, ZoomEarth can be seamlessly integrated with downstream models for tasks such as cloud removal, denoising, segmentation, and image editing through simple tool interfaces, demonstrating strong versatility and extensibility. All data and code will be released at https://earth-insights.github.io/ZoomEarth.

1. Introduction

Multimodal large language models (MLLMs) [5, 34, 36, 52] have been developing rapidly, demonstrating remarkable capabilities in multimodal understanding and perception. Several studies [18, 23, 42, 60] have further explored and applied vision-language models (VLMs) to RS images, achieving notable results in tasks such as visual question answering (VQA), grounding, and segmentation. However, satellite and aerial images often cover vast geographic areas, presenting significant challenges in efficiently inputting and processing UHR RS data.

*Equal contribution.
†Corresponding author (caoxiangyong@mail.xjtu.edu.cn)

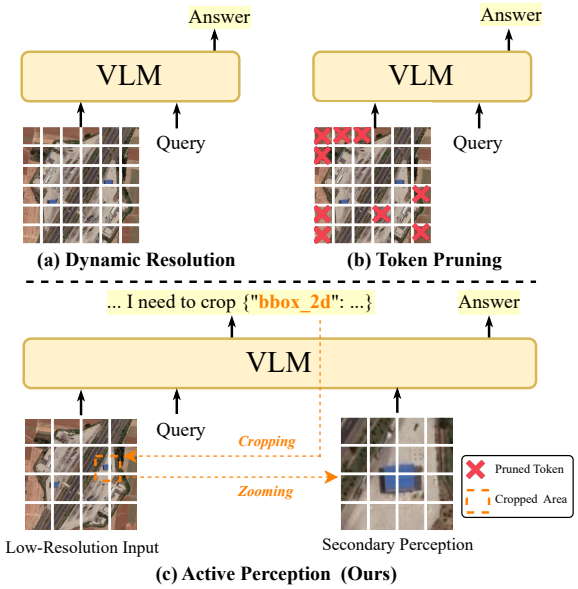


Figure 2. Comparison between passive perception and active perception. (a) Dynamic Resolution [8, 69] and (b) Token Pruning [2, 10] represent passive perception approaches with only a single image input. (c) We introduce a cropping–zooming based active perception method. "Zooming" refers to restoring the cropped image at its original high-resolution image.

To address this, dynamic-resolution-based methods [24, 35, 58, 71] are developed, enabling models to process high-resolution inputs directly but introduce significant computational cost. Token pruning methods [41, 54] enhance this process via manually crafted rules (*e.g.*, removing background tokens through clustering), yet they often fail to generalize to RS images with complex backgrounds. As shown in Fig. 2, these studies remain constrained to a passive perception paradigm that relies solely on a single visual input. Specifically, when attempting to acquire more fine-grained visual representations, the model requires higher-resolution image inputs, which in turn forces it to cope with redundant visual information interference.

In this work, to overcome the limitation introduced by passive perception, we explore a new active perception paradigm that enables models to revisit regions relevant to the given query. However, implementing active perception in UHR RS faces two key challenges: (1) the lack of RS image datasets that explicitly capture the active perception process, and (2) the difficulty of enabling models to perform adaptive region selection and active exploration. Recent studies [17, 57] mainly rely on optical character recognition (OCR) and object detection to construct supervised datasets, thereby achieving active perception through tool invocation. However, such approaches are not applicable to the RS domain, which typically lacks textual cues, and features targets that are often expansive regions (*i.e.*, the industrial zone) rather than solely discrete object instances.

To stimulate and enhance active perception in RS, we first present LRS-GRO, a large-scale benchmark dataset tailored for this paradigm. To explicitly model and supervise the active search process, the detail-oriented questions in LRS-GRO are annotated with precise bounding boxes (BBoxes) for the regions of interest (ROIs). In addition, for the supervised fine-tuning (SFT) dataset, each sample is accompanied by a Chain-of-Thought (CoT) that guides the model to first localize the ROI and then reason about the cropped details. To ensure the fidelity of these annotations, we introduce a novel semi-automatic annotation pipeline, which overcomes the defects observed in automated GPT-based workflows [41] on UHR RS images. Leveraging this pipeline, LRS-GRO encompasses 17 question types across global, region, and object levels, providing a comprehensive benchmark that evaluates not only VQA accuracy but also the active localization of correct ROIs.

Building on this foundation, we propose ZoomEarth, an adaptive cropping-zooming MLLM framework designed for active perception in UHR RS. The model first processes a downsampled global image to gain a holistic understanding. It then actively identifies ROIs, represented as BBoxes. These ROIs are subsequently cropped from the original high-resolution image and re-fed into the model for a detailed perception. As shown in Fig. 1, ZoomEarth can answer global queries from the initial view while adaptively "zooming-in" to handle region and object-specific queries.

The framework is trained using a two-stage strategy as shown in Fig. 3. First, SFT process teaches the model the basic localization-cropping-answering thought pattern. Second, we employ GRPO [46] to enhance its reasoning capabilities. Recognizing that standard IoU-based rewards are often too sparse for UHR images, we introduce a novel Region-Guided reward. This reward leverages the spatial distribution characteristics of RS scenes, providing denser and more fine-grained guidance by rewarding predictions based on their distance to the Ground Truth (GT).

Our main contributions are summarized as follows:

- We propose a UHR RS benchmark dataset, LRS-GRO, which encompasses multi-level question annotations and introduces an evaluation of active perception capabilities.
- We introduce ZoomEarth, a novel active perception framework that adaptively uses the cropping-zooming tool, and propose a Region-Guided reward tailored to the unique spatial properties of RS images.
- ZoomEarth achieves state-of-the-art (SOTA) on LRS-GRO and three public UHR RS datasets. It demonstrates strong potential as a foundation model for RS agents, integrating with downstream tasks like cloud removal, segmentation, and image editing in a training-free setting.

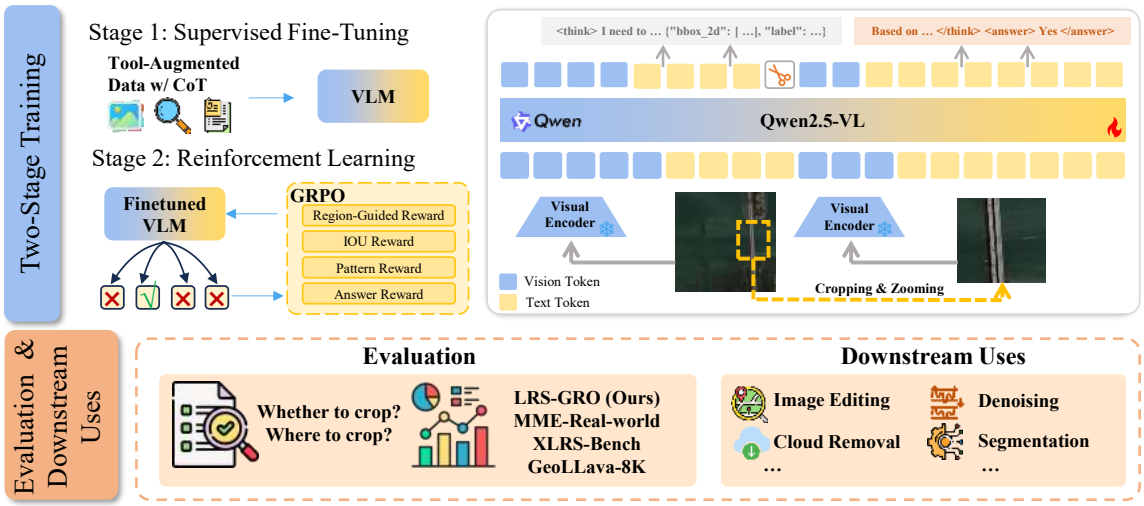


Figure 3. The visualization of the training and evaluation pipeline of our proposed methods. The model architecture diagram in the upper-right corner demonstrates the model's ability to adaptively crop the ROI by generating the BBox, and subsequently perform advanced reasoning. For clarity in the illustration, we omit the input of query tokens.

2. Related Work

2.1. High-Resolution Image Understanding

High-resolution image understanding persists one of the key challenges for Large Vision-Language Models (LVLMs) [21, 30, 48]. Some studies on LVLMs [18, 66] propose using positional embedding interpolation to increase input resolution. However, when the input size is further enlarged, the risk of excessive downsampling remains. Recently, advanced models have introduced dynamic-resolution strategies [4, 7, 32, 62], enabling native understanding of high-resolution images. Nevertheless, such approaches impose additional processing overhead and are susceptible to redundant visual information [20, 67]. In UHR RS imagery, small objects are significantly more numerous and densely distributed [50, 64], further highlighting this challenge. Some studies [11, 31, 40] attempt to introduce separate pathways to preprocess high-resolution image features. LRS-VQA [41] employs a Dynamic Image Pyramid to progressively crop local ROIs based on attention maps. GeoLLava-8k [9] adopts an adaptive token clustering strategy to remove background tokens. However, these models rely on manually predefined external rules and can effectively reduce redundant visual tokens only in specific scenarios (*e.g.*, bridge inspection with large waterbody backgrounds [26]). In this paper, we introduce the concept of active perception, enabling the VLM to obtain global context from entire RS imagery and autonomously acquire fine-grained local features, thereby enhancing its comprehension accuracy and reasoning efficiency.

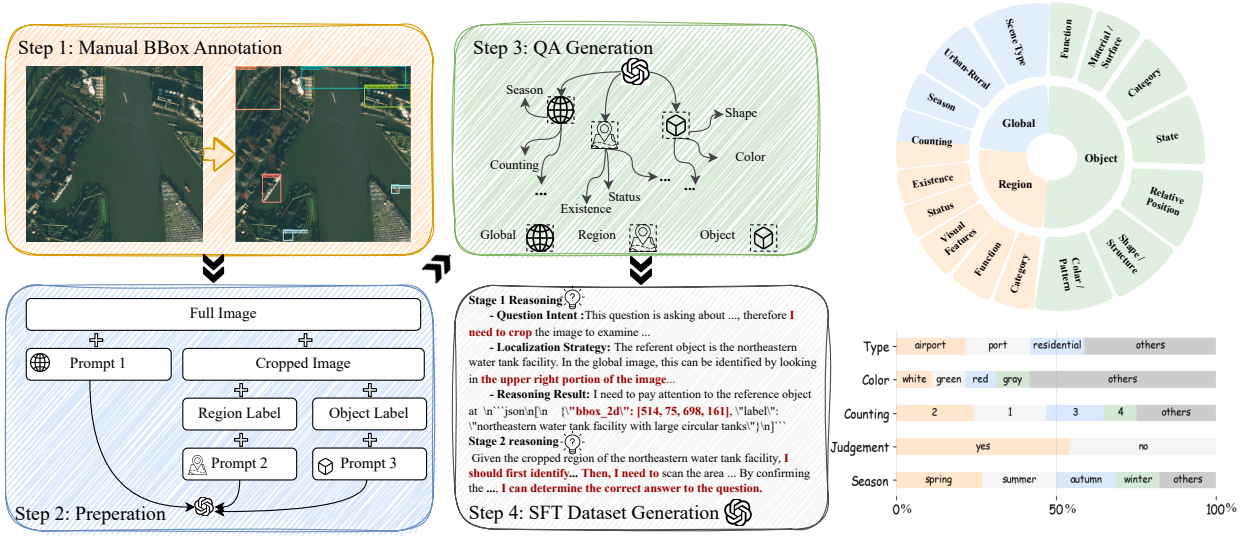
2.2. Visual Reasoning through COT

With the widespread application of CoT [6, 14, 28], VLMs have enhanced their visual perception abilities through reasoning, such as grounding tasks in images [38, 45]. However, reasoning in the textual modality can only rely on single-pass image perception, which entails the risk of missing critical visual information [53], leading to recent methods [49, 61, 70] that attempt to expand the perceptual boundaries of VLMs by leveraging external tools. For instance, Hu et al. [12] grants VLMs the ability to use tools for manipulating images, enabling them to enhance reasoning in mathematical and visual tasks by drawing auxiliary lines. Huang et al. [15] finds that models trained with reinforcement learning (RL) can effectively use appropriate tools for complex multimodal reasoning. Groundlight AI [19] introduces GRPO training, teaching model to use zoom-in tools for reasoning, which improves the ability to recognize small objects in natural images. These studies demonstrate that visual reasoning with tool invocation can significantly enhance the reasoning abilities of VLMs in complex tasks. However, VLMs with tools [49, 70] mainly rely on OCR and object detection to construct supervised data, which limits their applicability to RS imagery that lacks textual elements and contains densely distributed small objects. In this work, we introduce a localization process with varying region scales and a Region-Guided reward to enhance the model's capability for regional perception in RS images.

3. Method

3.1. Active Perception Oriented RS Dataset

Although recent studies [1, 56, 72] have emphasized the importance of benchmark datasets, resources specifically designed for high-resolution RS images remain scarce. Conventional RS VQA tasks typically operate on images of moderate resolution where objects are readily identifiable. In contrast, the UHR RS image, with its vast pixel space and



(a) Data Generation Pipeline (b) Data Categories and Proportion

Figure 4. (a) Construction pipeline of our proposed LRS-GRO dataset, in which manual filtering and refinement are performed after Step 3. (b) The upper chart shows the 17 question types in the LRS-GRO dataset, whereas the lower chart shows the distribution of typical answer categories, demonstrating the dataset's balance.

sparse target distributions, poses a substantial challenge for direct global observation. To address these challenges, we first propose a new benchmark task, Active Perception Oriented (APO) RS VQA, which requires models to actively locate, discriminate, and reason over fine-grained ROIs.

Definition 1 (Active Perception Oriented RS VQA)**.** *The **APO RS VQA** task evaluates two core capabilities: (i) accurately answering questions about UHR RS images; (ii) identifying and attending to image regions that are relevant to the given question.*

This task involves an iterative active perception process and requires a comprehensive evaluation of a model's semantic reasoning and spatial attention, assessing not only answer accuracy but also its active perception capability through ROI-based IoU.

3.1.1. Data Annotation Pipeline

To support this new task, we construct the LRS-GRO benchmark dataset using a novel semi-automatic annotation pipeline. Although LRS-VQA [41] has proposed a fully automated annotation method, it suffers from severe hallucinations of GPT when dealing with UHR RS images, making it unable to provide highly accurate BBox labels. To address this limitation, our process begins with manual annotation of BBoxes and their corresponding categories based on unique spatial characteristics (*e.g.*, "the top-most bridge" and "the building with a yellow roof"). Moreover, we explicitly distinguish between region-level and object-level BBoxes, providing hierarchical labels that better capture the unique properties of RS images. The Region-level BBoxes focus on planned land areas composed of multiple objects with distinct semantic labels, such as airports, residential communities, and industrial zones, and thus contain finer-grained compositional elements. In contrast, object-level BBoxes represent semantically homogeneous single objects, such as houses, airplanes, and ships.

Based on this, as shown in Fig. 4, we submit the entire image, cropped regions, and object-based enlarged crops to GPT-4o [44] to generate candidate Question–Answer (QA) pairs concerning attributes such as the target's color, shape, and status. In total, 40k candidate QA pairs are generated, which are then manually filtered and refined to retain the highest-quality 13k pairs while maintaining answer balance. For constructing the SFT dataset, we also instruct GPT-4o to produce step-by-step reasoning processes that incorporate the cropping procedure.

3.1.2. Dataset Composition and Structure

Following LRS-VQA [41], our LRS-GRO benchmark is curated from imagery sourced from FAIR1M-1.0 [51], GLH-Bridge [26], and STAR [27]. The final dataset comprises 1,224 high-resolution images (4,000–5,000 px), 3,592 bounding boxes, and 13,245 questions. Among them, 1,000 samples with detailed step-by-step CoT annotations constitute the SFT dataset.

We define 17 major question categories organized into 3 hierarchical spatial levels:

- **Global:** Counting, Season, Urban–Rural, Scene Type
- **Region:** Counting, Existence, Status, Visual Features, Function, Category
- **Object:** Function, Material / Surface, Category, State, Relative Position, Shape / Structure, Color / Pattern

For region and object-level tasks, we provide the refer-

Table 1. Comparison between RS vision-language benchmarks and our proposed dataset. "MME-R-W" denotes MME-Real-World [68]. The symbols ✓, ✗, and ✗represent manual, semi-automatic, and fully automatic annotation methods, respectively. Additional human verification is excluded.

Dataset	Avg. Resolution	Active Perception	VQA	
			Volume	Annotation
RSVQA [39]	512×512	N	111,134	✗
RSVQA-HR [39]	1,024×1,024	N	1,066,316	✗
VRSBench [25]	512×512	N	123,231	✗
RSIEval [13]	512×512	N	933	✓
XLRS-Bench [55]	8,500×8,500	N	32,389	✓
MME-R-W [68]	2000×1500	N	29429	✓
LRS-VQA [41]	5,000×5,000	N	7333	✗
LRS-GRO	5,000×5,000	Y	13245	✗

ence BBoxes to facilitate supervision and evaluation. This hierarchical structure is a key design choice, as it enables models to adaptively determine whether cropping is necessary based on the question's scope. Table 1 compares our dataset with existing benchmarks, underscoring its competitiveness in terms of resolution, scale, and its distinctive emphasis on active perception oriented VQA.

3.2. ZoomEarth

Building on the LRS-GRO benchmark, we propose ZoomEarth, an adaptive cropping-zooming framework designed for active perception. The core idea is to mimic human-like visual search by first gaining a holistic understanding from a coarse overview, then actively focusing on specific regions for detailed inspection. As shown in Fig. 3, ZoomEarth first processes a downsampled version of the entire UHR image to understand the global context. Subsequently, for questions requiring fine-grained detail, the model actively predicts a BBox for the relevant ROI, which is then cropped from the original high-resolution image and re-fed into the model for a detailed perception. To learn this complex behavior, we employ a two-stage training strategy.

3.3. Training via SFT and RL

3.3.1. Stage 1: Supervised Fine-Tuning

We first initialize the model through SFT. In this stage, the Qwen2.5-VL-3B [4] model is fine-tuned on our SFT dataset. This process enables the model to focus on RS domain knowledge, learn to follow the predefined formats for tool invocation. In addition, it enables the model to build a fundamental capability to distinguish among task levels and their respective localization requirements.

3.3.2. Stage 2: Reinforcement Learning

While SFT teaches imitation, we employ reinforcement learning to build a more robust and generalizable decision-making policy. We adapt GRPO [46] for this stage. Its critic-free design offers significant memory efficiency, making it particularly well-suited for handling the long visual token sequences generated from UHR images.



Figure 5. Comparison between r_{IoU} and r_{R-G}.

3.3.3. Reward Design for Active Perception

A significant challenge in the RL stage is designing an effective reward function. Previous research [17, 38, 47] on VLMs trained with RL often employ IoU as the primary reward. However, due to the foundational weakness of VLMs on UHR images, predicted BBoxes often deviate significantly from the GT, causing the IoU reward to remain zero and provide no learning signal as shown in Fig. 5.

To address this issue, we propose the Region-Guided reward, specifically designed for UHR RS images. This reward is inspired by the observation that geographic objects often exhibit strong spatial associations, such as aircraft located near airport terminals. Therefore, we argue that predictions located near the GT BBox should be rewarded. The Region-Guided reward provides a dense, fine-grained guidance signal by rewarding predictions based on their distance to the GT, as formulated below:

$$r_{R-G} = sigmoid(\frac{\alpha}{distance + \epsilon}) \quad (1)$$

where α denotes the scale number related to the resolution and distance denotes the Euclidean distance between the center of the predicted BBox and the GT BBox center, and ϵ is a small constant to prevent numerical overflow. The final reward is defined as:

$$Reward = r_{IoU} + r_{R-G} + r_{answer} + \beta r_{pattern} \quad (2)$$

where r_{IoU} represents the IoU reward, the same as in VLM-R^3 [17], r_{answer} is the normalized answer reward derived from word similarity [43], and r_{pattern} denotes the reward for the output format, assigned a value of 1 if correct and 0 if incorrect with the coefficient $\beta = 0.05$ controlling its influence.

3.4. Toolkit for Downstream Tasks

Through the preceding training stages, the model acquires the capability to adaptively crop ROIs, ranging from region-level to object-level. This introduces the potential of a foundation model tailored for a wide spectrum of downstream tasks in the RS domain. Considering that directly performing tasks such as denoising and clouding removal on high-resolution global images incurs substantial computational costs and often yields limited practical benefits. As the framework shown in Fig. 6, our cropping–zooming

method enables the model to more efficiently focus on user-specified ROIs in UHR RS. In a training-free manner, we integrate multiple advanced downstream models and design task-specific instructions to achieve adaptive tool invocation. Based on the ZoomEarth model, we construct a scalable RS agent capable of performing functions including segmentation, denoising, cloud removal, and image editing.

4. Experiment

4.1. Settings and Implementation Details

4.1.1. Dataset

Training Data. We utilize the LRS-GRO dataset, which is split into approximately 3,500 samples for training and 10,000 samples for testing. In the first stage, we fine-tune the model on the SFT dataset, with tool-necessary and tool-unnecessary subsets in a 2:1 ratio. In the second stage, we fine-tune the model on 2,500 samples using RL to enhance its tool utilization and generalization ability.

Test Data. For evaluation, we assess model performance on the LRS-GRO test set and other datasets with resolutions exceeding 5000×5000, including MME-Real-World [68], XLRS-Bench [55] and GeoLLava-8k [54].

4.1.2. Evaluation Metrics

Accuracy (Acc). We report Average Accuracy (Avg. Acc) for the VQA task across the aforementioned datasets. For LRS-GRO, the correctness of predictions is evaluated by measuring the semantic similarity between the predicted and ground-truth answers using WordNet [43], where a similarity score higher than 0.8 is regarded as correct[16, 41, 63]. For MME-Real-World [68], XLRS-Bench [55], we follow the evaluation setting provided by their respective datasets. Due to the lack of UHR RS datasets, we additionally incorporate the training dataset proposed by GeoLLava-8k[54] for testing, applying the same evaluation settings as that used for our proposed LRS-GRO dataset.

APO IoU. Standard VQA accuracy is insufficient as it does not evaluate how the model arrived at an answer. To measure the effectiveness of the localization step, we introduce the active perception oriented (APO) IoU. This metric is defined as the IoU between the predicted ROI bounding box and the ROI annotation. The APO IoU metric is applied to the region and object-level questions, as these tasks are designated to require tool invocation. A critical component of this evaluation is that if a model fails to generate a valid bounding box for a required task, its APO IoU for that sample is automatically set to 0. This directly penalizes failures in the active perception process, providing a comprehensive assessment of the model's tool-use capability.

4.1.3. Implementation Details

All experiments are conducted on 8 NVIDIA A800 GPUs. During training, the images are processed at an input resolution of 512 pixels (before cropping). In the RL stage, the sampling temperature is set to 0.7, while for all evaluation experiments, the temperature is fixed at 0.01. Regarding the learning rate, it is set to 3e-5 in the SFT stage and 1e-7 in the RL stage.

4.2. Main Results

Our model, built upon the Qwen2.5-VL 3B [4] architecture, consistently outperforms multiple baseline models on the LRS-GRO dataset, as shown in Table 2. Compared with InternVL3-8B that supports higher-resolution inputs, our model, with an initial input resolution of only 512, achieves substantial advantages in region-level and object-level tasks, improving accuracy by 1.53% and 4.00%. For global-level tasks that emphasize overall context rather than regional understanding, ZoomEarth is predictably lower than some larger models but still achieves a notable advantage over models of comparable size, such as Qwen2.5-VL-3B [4]. To ensure fairness, we adopt the Geochat [18] approach and fine-tuned its pre-trained weights on LRS-GRO using the same data. We observe that, despite Geochat having been trained on a large amount of additional RS data, ZoomEarth still demonstrates a significant advantage. Compared with the publicly available general-purpose tool-calling model VLM-R^3 [17], our smaller model achieves a 3.59% performance gain. Moreover, our APO IoU reaches 34.39%, which is substantially higher than VLM-R^3 [17]. This highlights the effectiveness and significance of our proposed dataset and training method for RS applications.

To evaluate the generalization capability of our model trained on the LRS-GRO dataset, we conduct zero-shot testing on three UHR RS datasets with all models evaluated under an input resolution of 1024. As shown in Table 3, compared with various VLMs from both general and RS domains, our model achieves SOTA performance on MME-RealWorld-RS [68], GeoLLava-8k [54], and XLRS-Bench [55]. Specifically, XLRS-Bench [55] provides predefined BBoxes in questions for region localization. Although this differs from our proposed framework that the model actively generates the BBox of the ROI, our method still demonstrates strong generalization ability.

4.3. Ablation Study

Ablation on cropping. To assess the effectiveness of the cropping–zooming tool, we conducted an ablation study by removing its application process. Specifically, we trained the model on the same dataset but required it to produce final answers without access to the cropped images. As shown in Table 4, the proposed active perception strategy yielded an average improvement of 4.39% across four

Table 2. Experimental results of various models on LRS-GRO. "Max Size" refers to the maximum resolution that the model can process as input. "method*" indicates we reproduce the SFT stage of existing methods using the same training data as ours. "-" denotes that the model lacks ROI localization outputs, making APO IoU evaluation impossible.

Leaderboard	LLM	Max Size	Global	Region	Object	Avg. Acc	APO IoU
LLaVA-OV-1.5 [3]	Qwen3-7B	2,304×2,304	62.43	36.18	39.92	45.33	-
IXC-2.5 [65]	InternLM2-7B	4,096×4,096	62.43	40.00	47.32	50.00	-
InternVL3 [59]	InternLM3-8B	3200×3200	**71.60**	44.58	47.80	53.67	-
Geochat [18]	Vicuna-1.5-7B	504×504	62.43	36.79	40.72	45.88	-
Qwen2.5-VL [4]	Qwen2.5-7B	1024×1024	69.39	38.47	43.58	49.62	-
		3333×3333	70.06	39.08	46.43	51.43	-
	Qwen2.5-3B	1024×1024	59.01	31.91	37.46	42.25	-
		3333×3333	58.90	31.76	38.66	42.83	-
VLM-R^3 [17] (w/ tools)	Qwen3-7B	512×512	69.72	44.83	37.40	50.17	19.93
Geochat*	Vicuna-1.5-7B	504×504	58.78	37.25	42.83	46.09	-
ZoomEarth (ours)	Qwen2.5-3B	512×512	63.09	**46.11**	**51.80**	**53.76**	**34.39**

Table 3. Zero-shot comparison of various models on MME-RealWorld-RS [68], XLRS-bench [55], GeoLLava-8k [54]. "–" indicates unavailable results due to potential data leakage risk from overlapping GeoLLava-8k and GeoChat training data.

Leaderboard	LLM	MME-realworld-RS	XLRS-bench	GeoLLava-8k	Avg. Acc
LLaVA-OV-1.5 [3]	Qwen3-7B	33.20	39.60	31.48	36.40
IXC-2.5 [65]	InternLM2-7B	36.12	31.50	36.20	34.61
InternVL3 [59]	InternLM3-8B	41.00	36.70	37.60	38.43
Geochat [18]	Vicuna-1.5-7B	28.62	22.03	-	-
Qwen2.5-VL [4]	Qwen2.5-3B	23.94	36.00	34.64	29.97
VLM-R^3 [17] (w/ tools)	Qwen3-7B	39.80	39.10	34.74	37.88
ZoomEarth (ours)	Qwen2.5-3B	**44.10**	**40.20**	**38.61**	**40.97**

Table 4. Ablation on cropping. "MME-R-W" means MME-RealWorld [68], XLRS refers to XLRS-bench [55]. '+ Cropping' indicates the incorporation of cropping–zooming functionalities.

Methods	LRS-GRO	MME-R-W	XLRS	GeoLLaVA-8k
ZoomEarth	51.10	42.10	30.00	34.30
+ Cropping	53.64↑2.57	44.10↑2.00	40.20↑10.20	38.61↑4.31

Table 5. Impact of RL on reasoning performance with tool usage.

Methods	Color	Count	Position	Overall
SFT	45.71	29.78	51.83	42.80
+ Cropping	45.40↓0.31	27.90↓1.88	49.58↓2.25	41.30↓1.50
RL	48.47	29.47	47.61	42.10
+ Cropping	49.69↑1.22	31.66↑2.19	50.14↑2.53	44.10↑2.00

Table 6. Ablation results for reward based on LRS-GRO.

r_{R-G}	r_{IOU}	Global	Region	Object	Avg. Acc
✗	✗	62.54	43.36	51.06	52.67
✓	✗	62.21	45.80	50.83	52.94
✗	✓	62.43	43.97	51.11	52.79
✓	✓	**63.09**	**46.11**	**51.80**	**53.76**

datasets. In addition, experiments conducted on other UHR RS datasets are under the zero-shot setting, which highlights the remarkable generalization ability of ZoomEarth.

Ablation on RL stage. To investigate the impact of the two-stage training strategy on enhancing the model's tool reasoning capability, we removed the RL stage on the LRS-GRO dataset. An ablation study was then conducted on the MME-RealWorld-RS dataset to avoid overfitting to the training set (LRS-GRO). As shown in Table 5, after SFT alone, invoking the cropping tool unexpectedly resulted in a performance drop. This finding suggests that SFT-based imitation learning primarily aids the model in acquiring output formats, whereas RL is crucial to the model's reasoning with tools and active perception capabilities.

Ablation on specific rewards. We conducted an ablation study on the LRS-GRO dataset to examine the effects of different rewards, focusing on the IOU reward and the Region-Guided reward. As shown in Table 6, removing the Region-Guided reward resulted in a performance drop of 0.97%, which is greater than the 0.82% drop observed when removing the IOU reward. This finding suggests that, for UHR RS images, the Region-Guided reward plays a more significant role than the IOU reward. Notably, compared with the setting without either IOU or Region-Guided rewards, adding the Region-Guided reward improved performance on region QA tasks by 2.44%, demonstrating stronger adaptability to region-level understanding questions.

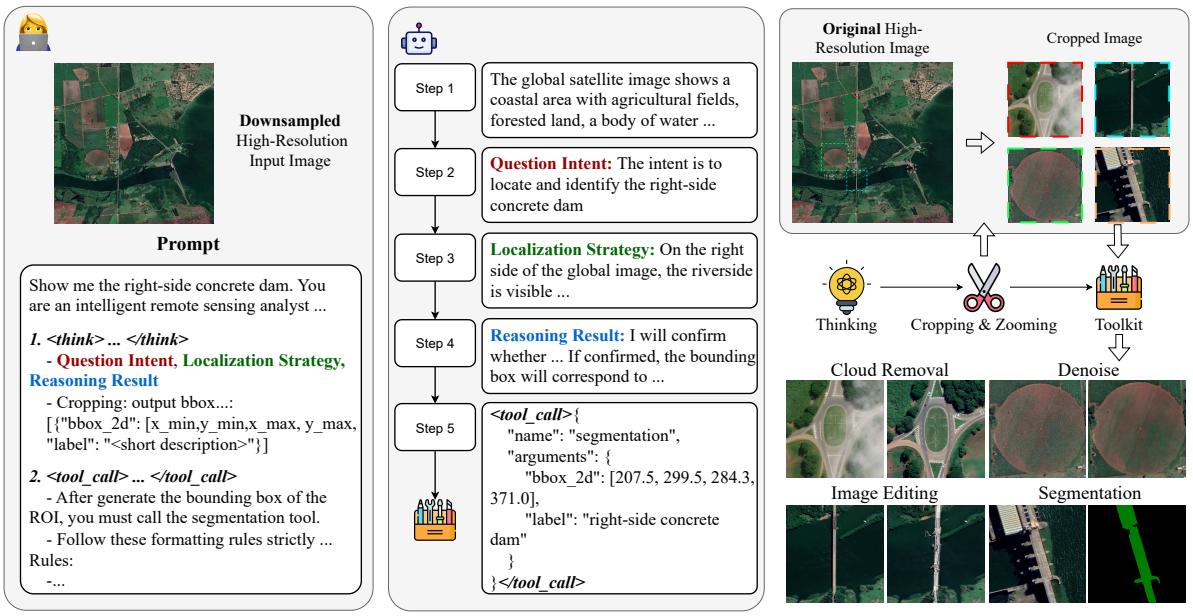


Figure 6. ZoomEarth inference pipeline with open tool invocation for cloud removal, denoising, image editing, and semantic segmentation. Due to the unavailability of suitable data for cloud removal, we simulated cloud-covered images using an image editing model [44].

Table 7. Ablation on inputting resolution. The model is evaluated during inference across various input resolutions under identical training data conditions.

Cropping	Max Size	Global	Region	Object	Avg. Acc	Speed (it/s)
✗	512×512	64.31	42.14	47.17	50.86	3.69
✗	1024×1024	63.76	41.37	48.32	51.16	3.50
✗	3333×3333	63.31	40.61	48.37	50.92	2.04
✓	512×512	**63.09**	**46.11**	**51.80**	**53.76**	2.38

Ablation on different resolution input. As shown in Table 7, we examine the performance differences of the model under varying input resolutions. The results indicate that, although increasing the resolution of input images provides more fine-grained visual information, it does not yield significant performance improvements, which means the model is easily disturbed by the redundant information introduced by a large number of visual tokens. However, after invoking the perception tool, the model still achieves a performance gain of over 2% at an input resolution of 512, suggesting that the our approach can effectively help the model overcome its performance bottleneck when processing high-resolution images and offers a significant speed advantage compared to 3333 × 3333 resolution input.

4.4. Unified Framework for Downstream Tasks

In practical applications, users typically focus on specific regions within high-resolution RS images, necessitating that the model possess the capability to autonomously identify and process the target regions. In our proposed approach, the model can identify ROIs using the cropping-zooming tool, and then performs subsequent processing reasoning. Through prompt-based instructions, we enable the model to invoke a broader set of tools and integrate specialized models from multiple domains, including cloud removal [37], denoising [29], image editing [33], and segmentation [22], as illustrated in Fig. 6. In implementation, we specify in the instructions that the model should output the BBox of the ROI within the tool invocation section, crop it from the original high-resolution image, and route it to the appropriate downstream model based on the determined tool name. Although our experiments are limited to a training-free setting, the results demonstrate that the model can adaptively select suitable tools from the toolkit according to the task requirements. The ZoomEarth model shows strong potential for UHR RS image processing and can be extended to a wide range of downstream tasks.

5. Conclusion

In this work, we introduce the concept of active perception to tackle the challenges of UHR RS VQA. We present LRS-GRO, a high-quality and diverse benchmark dataset spanning global, region, and object-level questions, enabling the stimulation and evaluation of active perception capabilities. Our proposed ZoomEarth framework adaptively calls the cropping–zooming tool to actively explore and extract fine-grained information from UHR images, guided by our proposed reward mechanism tailored to the distributional characteristics of RS scenes. Experimental results demonstrate competitive performance, strong generalization, and a scalable foundation for integrating downstream tasks. We anticipate that ZoomEarth will form the foundation for designing future autonomous RS agents, equipping them with the fundamental capability to actively seek information within complex, large-scale environments.

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